Trapezoidal integration for the prediction step

- For a 1-d problem, we can write the prediction step as:

$$\begin{aligned} f(x_k \mid \mathbb{Z}_{k-1}) &= \int_{x_{\min}}^{x_{\max}} f(x_k \mid x_{k-1}) f(x_{k-1} \mid \mathbb{Z}_{k-1})\,\mathrm{d}x_{k-1} \\ &= \int_{x_{\min}}^{x_{\max}} g(x_{k-1}; x_k, \mathbb{Z}_{k-1})\,\mathrm{d}x_{k-1} \\ &\approx w\left[\frac{g(x_{\min}; x_k, \mathbb{Z}_{k-1}) + g(x_{\max}; x_k, \mathbb{Z}_{k-1})}{2} + \sum_{i=1}^{N_r-1} g(x_{\min} + iw; x_k, \mathbb{Z}_{k-1})\right]. \end{aligned}$$

- We are now interested in implementing this operation numerically.
  - We can't implement the integral over an infinite set of values of $\mathrm{d}x_{k-1}$, so we must discretize the dummy variable of integration x_{k-1} over N_r regions (3rd line of eqn.).
  - Similarly, we cannot store all values of $f(x_k \mid \mathbb{Z}_{k-1})$ corresponding to an infinite set of x_k. Need to discretize $f(x_k \mid \mathbb{Z}_{k-1})$ by discretizing x_k into N_r regions.

Tradeoff in precision versus speed

- If we discretize x_k into N_r regions, then we must compute:

$$\begin{aligned} f(x_k \mid \mathbb{Z}_{k-1}) \approx w\Bigg[&\frac{g(x_{\min}; x_k, \mathbb{Z}_{k-1}) + g(x_{\max}; x_k, \mathbb{Z}_{k-1})}{2} \\ &+ \sum_{i=1}^{N_r-1} g(x_{\min} + iw; x_k, \mathbb{Z}_{k-1})\Bigg]. \end{aligned}$$

  for each of those N_r possible values of x_k.
- We have N_r additions for each of the N_r values of x_k: $\mathcal{O}(N_r^2)$ operations overall.
  - "Curse of dimensionality" when dealing with multi-dimensional problems: $\mathcal{O}(N_r^{2n})$.
- The limiting factor to computing an estimate of $f(x_k \mid \mathbb{Z}_k)$ is precision vs. speed:
  - We increase precision by increasing N_r;
  - We increase speed by decreasing N_r.
- Brute-force integration not an appealing final solution, but gives high-quality results.

Tracking example introduction

- In the next lesson, we will look at Octave code to implement a 1-d Bayesian recursion.
- For now, we set up a sample problem to solve: Suppose that we wish to track time-varying x_k for a system modeled by:

$$\begin{aligned} x_{k+1} &= \alpha x_k + w_k \\ z_k &= 0.1x_k^3 + v_k, \end{aligned}$$

  where $w_k \sim \mathcal{N}(0, \sigma_w^2)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$ are mutually uncorrelated and IID.
- When running a simulation, we need to select values for design parameters:
  - Number of simulation iterations: We will use $k = 0 \ldots 200$.
  - Range of integration: We will use $x_{\min} = -3$ and $x_{\max} = 3$.
  - Number of regions for x_k: We will use $N_r = 500$ regions.
  - We will further define model parameters: $\sigma_w = 0.2$, $\sigma_v = 0.1$, and $\alpha = 0.99$.

Tracking example prediction step

- For the prediction step, numerically integrate to approximate:

$$f(x_k \mid \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k \mid x_{k-1}) f(x_{k-1} \mid \mathbb{Z}_{k-1}) \, \mathrm{d}x_{k-1}.$$

- The pdf $f(x_{k-1} \mid \mathbb{Z}_{k-1})$ is the output from the prior iteration: a vector of N_r values.
- We determine the pdf $f(x_k \mid x_{k-1})$ from the model state equation.
  - Since $x_k = \alpha x_{k-1} + w_{k-1}$, if x_{k-1} is considered to be known, x_k is the sum of a known constant and a Gaussian random variable $w_{k-1} \sim \mathcal{N}(0, \sigma_w^2)$.
  - Therefore, if x_{k-1} is known, x_k is Gaussian and its conditional pdf is determined by its mean and variance. $\mathbb{E}[x_k \mid x_{k-1}] = \alpha x_{k-1}$ and var$(x_k \mid x_{k-1}) = \sigma_w^2$.
- So, $f(x_k \mid x_{k-1}) \sim \mathcal{N}(\alpha x_{k-1}, \sigma_w^2)$. Can evaluate for any value of x_k via:

$$f(x_k \mid x_{k-1}) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left(-\frac{(x_k - \alpha x_{k-1})^2}{2\sigma_w^2}\right).$$

Tracking example update step

- For the time-update step, we directly compute:

$$f(x_k \mid \mathbb{Z}_k) = \frac{f(z_k \mid x_k) f(x_k \mid \mathbb{Z}_{k-1})}{f(z_k \mid \mathbb{Z}_{k-1})}.$$

- The pdf $f(x_k \mid \mathbb{Z}_{k-1})$ is the output of the prediction step, an N_r-element vector.
- We determine the pdf $f(z_k \mid x_k)$ from the model output equation.
  - Since $z_k = 0.1x_k^3 + v_k$, if x_k is considered to be known, z_k is the sum of a known constant and a Gaussian random variable $v_k \sim \mathcal{N}(0, \sigma_v^2)$.
  - Therefore, if x_k is known, z_k is Gaussian and its conditional pdf is determined by its mean and variance. $\mathbb{E}[z_k \mid x_k] = 0.1x_k^3$ and var$(z_k \mid x_k) = \sigma_v^2$.
- So, $f(z_k \mid x_k) \sim \mathcal{N}(0.1x_k^3, \sigma_v^2)$. Can evaluate for any value of z_k via:

$$f(z_k \mid x_k) = \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{\left(z_k - 0.1x_k^3\right)^2}{2\sigma_v^2}\right).$$

Tracking example additional details

- For the time-update step, note the denominator of:

$$f(x_k \mid \mathbb{Z}_k) = \frac{f(z_k \mid x_k) f(x_k \mid \mathbb{Z}_{k-1})}{f(z_k \mid \mathbb{Z}_{k-1})}.$$

  - We can compute this conditional pdf via the law of total probability (a giant pain).
  - Or, we can recognize that its only function is to ensure $f(x_k \mid \mathbb{Z}_k)$ integrates to 1.
  - It is easiest to compute the numerator first, then divide that computation by its integral to normalize the pdf $f(x_k \mid \mathbb{Z}_k)$ to have an integral of one.
- Finally, note that the predict/update recursion requires that $f(x_0 \mid \mathbb{Z}_0)$ be specified, where:

$$f(x_0 \mid \mathbb{Z}_0) = \frac{f(z_0 \mid x_0) f(x_0 \mid \mathbb{Z}_{-1})}{f(z_0 \mid \mathbb{Z}_{-1})}.$$

  - Since $\mathbb{Z}_{-1}$ is unknown, we assume that $f(x_0 \mid \mathbb{Z}_{-1}) = f(x_0) \sim \mathcal{N}(0, \sigma_w^2)$.

Summary

- In this lesson, you learned how to evaluate sub-components of the Bayesian recursion, which were abstract until this point.
- You have seen more details on how to compute the integral in the prediction step that estimates $f(x_k \mid \mathbb{Z}_{k-1})$.
  - This included an introduction to the "curse of dimensionality" that the particle filter will need to overcome eventually.
- You have seen how to derive and compute $f(x_k \mid x_{k-1})$ and $f(z_k \mid x_k)$.
- We are now ready to implement this example in Octave code to see the kinds of results for $f(x_k \mid \mathbb{Z}_k)$ we might expect.
  - Once we have $f(x_k \mid \mathbb{Z}_k)$, we can compute mean, median, mode, whatever.
  - Can plot these point estimates as functions of time, just like with a Kalman filter.